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# Fog Removal Method of Slope Monitoring Image Based On Vision Detection

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**Abstract:**In the slope monitoring based on image detection, the main work is to process the acquired slope image. The landslide occurred mostly in the rain, fog and other complex weather conditions. If we can process effectively and fast fog images according to the fog horizon slope vision images. It would be helpful for subsequent image segmentation, object extraction, positioning, and improving the accuracy and efficiency of detection of slope deformation. Based on the visual technology in slope monitoring, we compared two kinds of defogging algorithm of slope images. One is a kind of image enhancement method of non-physical model, mainly including: equalization algorithm and homomorphic filtering algorithm, McCann Retinex algorithm and multi-scale Retinex algorithm and a global histogram. The other is the image restoration method based on physical model, including the dark channel prior bilateral filtering algorithm, and combined with the theory of dark channel prior to fog algorithm. The experimental results show that the histogram equalization method has the advantage of fast imaging quality in slope visual image processing, and is more suitable for slope monitoring.

**Key Words:** Slope monitoring, Image processing, Dark Channel Prior, Bilateral filtering

## 1 Introduction

China is a mountainous country and its mountainous area accounts for about 2/3 of the total area. It is one of the world's landslide prone countries, especially in the southwest and Northwest Mountainous Area. Landslides can destroy the local roads, railways and other infrastructure, cause traffic interruptions, bury villages, seriously affect traffic safety, people's life and property safety. The slope is the most potential to geological disasters, geological structure, is geologically loose due to being unstable, and can easily lead to natural disasters such as landslide and collapse in the long-term accumulation of slope deformation. It is necessary to strengthen the monitoring of the slope that may be security risks, realize the early warning of danger, and avoid the occurrence of corresponding safety accidents.

Now there are many different kinds of methods of slope monitoring, mainly including artificial direct observation method [1], GPS detection method [2], laser 3D imaging [3], synthetic aperture radar measurement [4], acoustic emission monitoring method [5] and image detection method [6]. Image detection is a method of measuring target slop displacement using image acquisition equipment as data source. Image detection has characteristics of non-contact, high accuracy. The method can dynamically reflect slope deformation. The core of the method is to acquire the image of the slope, get the coordinates of the target through the image, and then do coordinate transformation and calculate the displacement of the slope. The slope displacement monitoring by monocular vision only needs one camera to get the slope image, and the displacement and deformation of the slope can be obtained by image processing.

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Because of the characteristics of wet weather and foggy weather in southwest area, the target feature is not obvious in monocular vision acquisition, which is more difficult for subsequent operation. Image processing is very important for detection of slope displacement. The method of fog removal of image is divided into two types: image enhancement and image restoration. Therefore, the combination of image fog removal algorithm and visual image processing of slope monitoring is helpful for the operation of slope image in foggy weather, and the function of slope monitoring is better achieved. In this paper, several algorithms of image fog removal will be introduced.

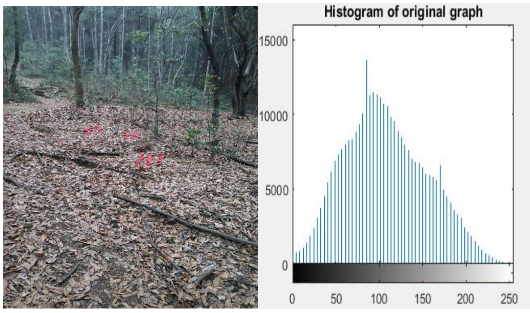
## 2 Algorithm of image fog removal

At present, there are many algorithm of image fog removal, which are divided into two categories: image enhancement based on non-physical model and image restoration based on physical model. The image enhancement method does not need to consider the reason of image degradation, only from the fog image with low contrast and fuzzy characteristics, through enhancing the contrast of the image. The detail information is more prominent, while the removal of redundant information or attenuation achieves the effect of the image fog removal. Image restoration method analyzes and analyzes the atmospheric scattering effect by analyzing the degraded mechanism of foggy image, designs and builds foggy image degradation model, realizes the restoration of scene in image, and gets clear fog image.

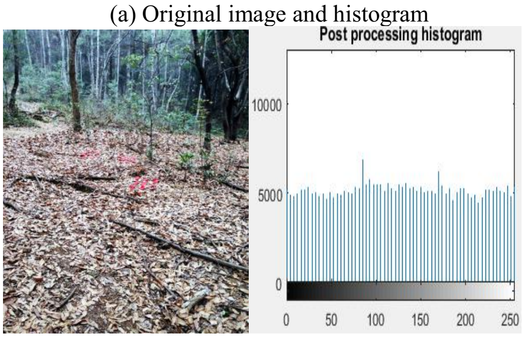
### 2.1 Image enhancement method based on non-physical model

The technology of image enhancement fog is not analyzing the reason of image degradation, but based on different needs, highlighting or improving some characteristic information of image, improving the contrast of image, and achieving the purpose of image fog removal. Using image enhancement to deal with foggy images is in essence the enhancement of image contrast. The main image enhancement methods include histogram equalization algorithm [7], homomorphic filtering algorithm [8], Qu Bo transformation, wavelet method, and Retinex algorithm [9] and so on. Here are several common image enhancement methods.

In the image, each pixel has its corresponding gray value, and the number of pixels corresponding to each gray value in the statistical image is counted, and the histogram drawn is called the gray histogram. The purpose of histogram global equalization algorithm is to focus on the grayscale of a certain area and stretch it evenly to each gray level, so that the range of the gray value of the image is increased, the overall contrast of the image is enhanced, and the fog removal effect is achieved. Therefore, the histogram global equalization algorithm is suitable for processing the whole dark or bright image on the whole. It is not suitable for processing images with large differences in local luminance, which will reduce the gray level of the original image. The processing results are shown in Fig 1.




(a) Original image and histogram




(b) Post processing image and histogram

Fig. 1: Processing results of histogram global equalization algorithm

From Fig 1, we can see that the original image is mist. The histogram pixels are mainly concentrated in the gray value of 50~200. After processing, pixels in the image histogram are evenly distributed in all the gray values, and the resolution of the images is higher.

Homomorphic filtering algorithm is a frequency domain based on image enhancement algorithm. It combines image grayscale transformation with frequency filtering based on the image's illumination and reflection model, improves image contrast, enhances image details, and makes the processed image clearer.

Homomorphic filtering is the form of the image f(x,y) as the product of the radiation component i(x,y) and the reflection component r(x,y), and the expression is as follows:

$$f(x,y) = i(x,y)*r(x,y) \tag{1}$$

The radiation component i(x,y) is mainly determined by the intensity of illumination, corresponding to the corresponding low frequency components. The reflection component r(x,y) mainly reflects the detailed information of objects in the image, corresponding to the high-frequency components. Therefore, the appropriate filter can be selected to deal with the high frequency and low frequency components in the image, and a clear map image can be obtained. For example, when the image of uneven illumination, homomorphic filtering algorithm, reduce the low-frequency components in the image and increase the high frequency components, which reduce the irradiation components in the image enhanced reflection component, the intensity of uneven impact on the image is weakened and image details are enhanced.

(a) Original image (b) Post processing image

Fig. 2: Processing results of homomorphic filtering algorithm

From Fig 2, the image sharpness of the image processed by homomorphic filtering is enhanced, and the details in the distance are also improved.

Retinex theory is proposed by Edwin H Land and McCann based on color constancy. Retinex theory holds that human perception of object surface color is mainly determined by the reflection properties of the object surface, but not related to the illumination intensity of the external environment.

The Retinex theory regards the image F(x,y) as the product of the reflection component R(x,y), determined by the environmental factors, I(x,y) and the essential attribute of the object, as follows:

$$F(x,y) = I(x,y)*R(x,y) \tag{2}$$

The radiation component I(x,y) determines the dynamic range of the image F(x,y), and the reflection component R(x,y) reflects the reflection property of the object itself. The essence of Retinex algorithm is to regard image as the sum of radiation components and reflection components, then remove the radiation components, eliminate the influence of external conditions on images, and get the reflection component, that is the original property of an object.

The McCann Retinex algorithm uses multiple iterative methods to estimate the brightness of the image by comparing the light and dark relations between pixels in the image of the image.

(a) Original image (b) Post processing image

Fig. 3: Processing results of McCann Retinex algorithm

From Fig 3, we can see that after McCann Retinex algorithm, the color fidelity of the processed image is better, and there is no obvious phenomenon of color distortion. As a whole, the effect of fog removal is general, the image is still shrouded in mist, and the information resolution of local area is not high.

When the image is F(x,y), the radiation component of the ambient light is I(x,y), and the reflection component of the target object is R(x,y), the single scale Retinex algorithm can be expressed as:

$$\log R(x,y) = \log F(x,y) - \log I(x,y) \quad (3)$$

Assuming that the reflection of light and image tends to be more smooth than that of the object surface, we can make the environmental luminance image I(x,y) into the convolution form of the surrounding function and the corresponding channel light intensity, and the reflective component can be expressed as:

$$r_i(x,y) = \log F_i(x,y) - \log [G(x,y)*F_i(x,y)] \quad (4)$$

Among them, Fi (x, y) represents the brightness of the environment, that is, the I color channel of the input image. Ri (x, y) represents Retinex output, and G (x, y) is a lowpass convolution function. Here we use Gauss form to estimate the luminance image from the original image. The expression is:

$$G(x,y) = \lambda e^{-(x^2+y^2)/c^2} \quad (5)$$

Where c is the scale factor Gauss surround function, the smaller the c, the image compression is larger, the image shadow details are displayed more clearly, but the output color easily distorted; if c is larger, the better the overall visual effect after image enhancement, but the details of the image display is not clear.

To achieve the balance between gray scale dynamic range compression and contrast enhancement, an appropriate scaling factor must be selected. After many experiments, it is found that the general value is 50-100. The value of the λ is determined by the normalized function.

$$\iint G(x,y)dxdy = 1 \quad (6)$$

The multi-scale Retinex algorithm is based on weighted average superposition of multiple single scale Retinex algorithm. The reflection component of incident light is obtained through convolution of Gauss function and original image. The concrete operation formula is as follows:

$$R(x,y)=\sum_{i=1}^{N} w_i \left\{\log F_i\ (x,y) - \log[G_i(x,y)*F_i(x,y)]\right\} \quad (7)$$

Among them, N shows the number of single scale selections. In general, N value is 3; Wi represents weighted coefficient; Gi represents different scale Gauss surround function, and its expression is:

$$G_i(x,y) = \lambda e^{-(x^2+y^2)/c_i^2} \quad (8)$$

(a) Original image (b) Post processing image

Fig. 4: Processing results of multi-scale Retinex algorithm

From Fig 4, the image contrast and resolution multi-scale Retinex algorithm after treatment are very high, the local image details clear, identifiable degree is high; the original scene information of image well preserved; but after processing the image color distortion is a serious partial color.

### 2.2 Image restoration based on physical model

The model based on image defogging method is also called image restoration. This kind of method is mostly based on the atmospheric scattering model proposed in McCartney [10], introduces the principle of image degradation, so long as the fog image reverse processing the image due to the degradation and distortion of the part to be compensated, obtains a restored image. Before the application of this kind of method, some prior knowledge and theory should be obtained as the basis of image restoration.

In 2009, Kaiming He et al, put forward the dark channel prior theory [11], which is a statistical rule for the outdoors foggy images. Kaiming He and others through no fog image processing of a lot of outdoor images, found an objective statistical law, if there are any patches in the vast majorities of outdoor fog image, there are always some or at least one pixel, and some color channel strength value is very low, and close to zero, called the dark channel.

From the McCartney atmospheric scattering model, the fog day image degradation model is as follows:

$$I(x) = J(x)t(x) + A(1 - t(x)) \quad (9)$$

Among them, x represents the position coordinates of each pixel in the image, I(x) represents objects reflected light after atmospheric attenuation to the light intensity of the imaging equipment, which has obtained the fog image, J(x) represented by no fog image after image processing, A represents the atmospheric optical components, usually assumed to be global constants, t(x) on behalf of the transmission, transmission means the light reflected from the surface of the object through the fog into the number of imaging devices.

Kaiming He proposed the use of dark original color theory to solve transmittance t(x) and atmospheric light A.

First, the fog image is divided into 15*15 blocks, and the local and global dark original color images are obtained. Then, the dark original color image is used to estimate the transmittance and atmospheric light. Method of using soft matting to refine the transmittance, the transmittance of t(x), A and atmospheric light fog image I(x) into formula (9), obtained after image restoration with J(x). Fig 5 is a flow chart of the dark channel prior algorithm:

To estimate the value of the atmospheric light A, first, the first 0.1% pixels are taken from the dark original color map based on the size of the brightness, most of which are opaque. Then in the above pixel points, the value of the maximum pixel point in the original fog image I(x) and its corresponding intensity is A of the atmospheric light.

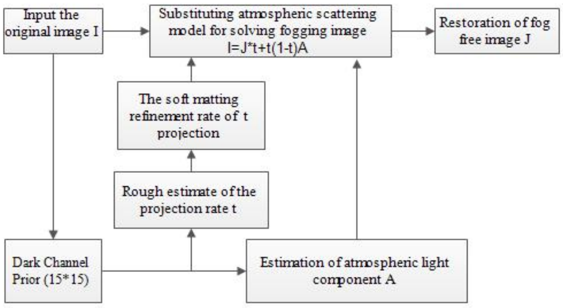



Fig. 5: Dark channel prior algorithm flow chart

From the dark channel prior theory and the image degradation model of the McCartney atmospheric scattering model, the transmittance of t(x) can be estimated. It is assumed that the atmospheric light component A is known, and the A value is constant greater than zero, and formula (10) can be expressed as:

$$\frac{I^c(y)}{A^c} = t(x)\frac{I^c(y)}{A^c} + (1 - t(x)) \tag{10}$$

At the same time, the minimum value of the local area Ω(X) of the R, G and B three channels is filtered simultaneously.

$$\min_{y\in\Omega(x)}(\min_c \frac{I^c(y)}{A^c}) = t(x)\min_{y\in\Omega(x)}(\min_c \frac{I^c(y)}{A^c}) + (1 - t(x)) \tag{11}$$

It is known from the knowledge of dark channel prior theory that when the image is a no foggy image, the value of dark original color is about 0, that is:

$$J^{dark}(x) = \min_{y\in\Omega(x)}(\min_c J^c(y)) = 0 \tag{12}$$

Due to the positive number of $A_c$, the expression of the type (12) into (11) to get the transmittance t (x) is as follows:

$$t(x) = 1 - \min_{y\in\Omega(x)}(\min_c \frac{I^c(y)}{A^c}) \tag{13}$$

This can be a rough estimate of the transmission rate. However, because of the previous use of the block statistics dark original color, there is a massive effect between the local blocks, so it is necessary to refine and smooth the transmissivity picture.

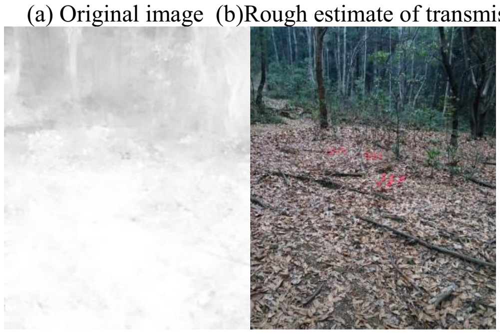
Kaiming He found that the atmospheric scattering model formula (9) and image matting equation (14) in the form of expression is consistent with [12]. The image matting is a foreground and background separation technology, and its expression is:

$$I = F\alpha + B(1 - \alpha) \tag{14}$$

Among them, I and F and B represent synthetic images, foreground images and background images. α is a mask that indicates the degree of opacity of pixels, changing continuously between 0 and 1. When α=1, the foreground is completely opaque and the background is covered up. When α=0, the foreground is completely transparent, and the background can be clearly seen.

By observing the atmospheric scattering model, it can be seen that the transmittance diagram can be considered as an α graph. The process of calculating transmission graph can be seen as for the continuous change of α matting algorithm in graph. In summary, the refinement of transmittance graph cutting method, is to get the image of a scene object contour generally clear, relatively distinct levels of scene depth. Therefore, Kaiming He used to fine transmissivity diagram by matting algorithm.

(a) Original image (b)Rough estimate of transmissivity



(c)Figure after the transmittance of soft matting (d)Post processing image

Fig. 6: Processing result of dark channel prior algorithm

From Fig 6 is can be seen that the contrast and resolution of the foggy image processed by the dark channel prior algorithm are all high. From the transmission map, with a soft matting method, the transmission can be refined and can improve the original block effect in transmission

Bilateral filtering is a nonlinear filtering method [13]. It is similar to Gauss filter, and it is also the principle of using local weighted average, but there are obvious differences between the two. The weighting coefficient of bilateral filtering is determined by two parts. One is the difference between the spatial distance between pixels, which can be called spatial proximity factor, and the other is the difference between gray values of pixels, which can be called grayscale similarity factor. In addition, bilateral filtering is more complex in principle. Bilateral filtering can not only smooth the input image, but also effectively maintain the edge information of the image. After bilateral filtering is applied to image processing, the gray value of each pixel in the image is equal to the weighted average value of the pixels in the neighboring area, and the weighting coefficient is equal to the product of the gray similarity factor and the spatial proximity factor [14]. Therefore, the combination of bilateral filtering and dark original color prior knowledge can be used in the fog image processing, and the better processing results can be obtained. The steps of the algorithm are as follows:

1、Using the smoothing and edge preserving characteristics of bilateral filter, we estimate the initial value of the atmospheric scattering function and show the detail information such as the contrast of the local area of the image [15,16].

2、The dark channel prior method proposed by Kaiming He is used to estimate the atmospheric light value A.

3、The atmospheric scattering function and atmospheric light value acquired by steps 1 and 2 are used to obtain the atmospheric transmittance, and then the image is restored and the restored image is output.

(a) Original image (b) Post processing image

Fig. 7: Processing results of bilateral filtering and dark channel prior combination algorithm

From Fig 7, we can see that the fog removal effect of this algorithm and He Kaiming He algorithm is better, which makes the image sharpness and the target area clearer, but the whole image is slightly distorted.

## 3 Experimental results and analysis

Subjective evaluation and objective evaluation are often used in the analysis of the effect of various algorithms on the slope image removal. Subjective evaluation is the analysis of the processing effect of the tested slope image based on human's subjective feelings. It has been analyzed in the previous processing results, and is no longer described here. The objective evaluation is the perception mechanism simulation of human vision system to measure the quality of the image, in order to analyze the quality of the image to fog after the slope, by contrast, information entropy, average gradient, mean and standard deviation of evaluation, the evaluation index calculation results are shown in Table 1 and in Table 2.

The contrast reflects the overall distribution of gray and black and white contrast. The greater the contrast in the image contrast of black and white, more obvious, the better image clarity, from table 1 that the global histogram balance method and multi-scale Retinex method to contrast the best image fog removal effect, homomorphic filtering and bilateral filtering method to contrast the worst image fog removal effect.

Table 1: Contrast, average gradient and information entropy comparison of the processing results of each algorithm

| | contrast | average gradient | information entropy |
|---|---|---|---|
| original image | 37.941 | 20.289 | 7.597 |
| histogram global equilibrium | 63.457 | 31.519 | 7.951 |
| homomorphic filtering | 27.002 | 16.204 | 6.844 |
| McCann Retinex | 42.504 | 24.977 | 7.673 |
| Multiscale Retinex | 66.907 | 37.255 | 7.625 |
| Kaiming He algorithm | 41.785 | 21.623 | 7.665 |
| Bilaterally filtered dark channel method | 25.692 | 17.173 | 7.109 |

The average gradient reflects the image contrast on the expression of tiny details, used to evaluate the image clarity, from table 1, that the global histogram balance method and multi-scale Retinex method processing result of image definition best, homomorphic filtering and bilateral filtering method of dark channel image processing results clarity is poor.

Information entropy is the average amount of information of the image, reflecting the amount of information of the image. From table 1, it is known that the amount of information processed by each algorithm is roughly equivalent.

Table 2: Comparison of the mean value, standard deviation and processing time of each algorithm

| | Mean value | Standard deviation | processing time /s |
|---|---|---|---|
| original image | 107.69 | 48.22 | \ |
| histogram global equilibrium | 127.46 | 74.442 | 0.453 |
| homomorphic filtering | 44.188 | 39.553 | 2.077 |
| McCann Retinex | 163.83 | 54.296 | 5.949 |
| Multiscale Retinex | 106.61 | 77.859 | 3.962 |
| Kaiming He algorithm | 98.287 | 52.504 | 130.273 |
| Bilaterally filtered dark channel method | 56.266 | 45.921 | 33.034 |

The mean value refers to the average pixel value of the image. It reflects the average image brightness, average brightness is bigger and the image quality is better. From table 2 is can be seen that the global histogram equilibrium method and McCann method of Retinex image processing results best quality, while homomorphic filtering and bilateral filtering method for image processing results of dark channel quality is poor.

Standard deviation refers to the degree of discreteness of the pixel gray value of the image relative to the mean value. If the standard deviation is bigger, it shows that the more grayer level the image, the better the image quality. From table 2, we know that the histogram global balance method and multi-scale Retinex method have the best image quality.

In the image processing time, the histogram global balance method is the least, the Kaiming He algorithm is the longest time and the processing speed is the slowest.

## 4 Conclusion

In this paper, the fog removal method of visual image in slope monitoring is studied. Mainly for the non-physical model image enhancement method and for the physical model image restoration method of two kinds of image fog removal algorithm based on the analysis of, several better fog removal algorithms are compared and analyzed. The experimental results show that the visual, global histogram equalization algorithm and Kaiming He algorithm on slope image fog removal effect best, bilateral filtering method based on dark channel in some detail areas of good fog removal effect, but the overall image appeared as a serious color. Combined with the analysis of image performance indicators after each algorithm, histogram global balance method has the characteristics of fast imaging quality in slope visual image processing, and is more suitable for slope monitoring applications.

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